

EXPLORING GLOBAL UNIVERSITY EXCELLENCE

An Interactive Visualization of
QS World University Rankings

MILESTONE 3

PROCESS BOOK

GLOBAL DISTRIBUTION

Number of Ranked Universities by Country

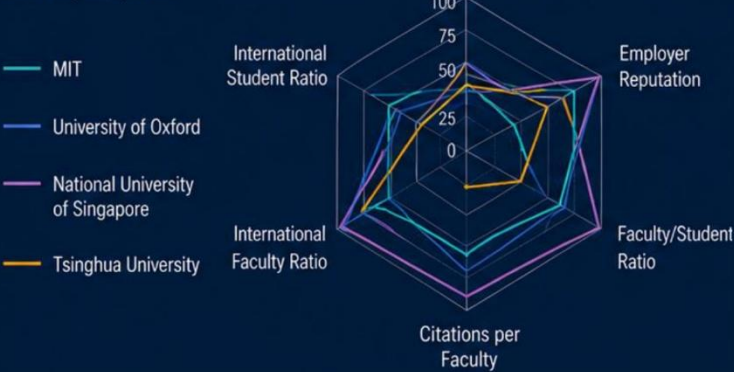


ACADEMIC REPUTATION vs OVERALL SCORE



INDICATOR PROFILE

University Comparison



RANKING TABLE

Filtered by: All Countries

Rank	University	Overall Score	Academic Reputation	Employer Reputation	Citations per Faculty
1	Massachusetts Institute of Technology	100.0	100.0	100.0	99.9
2	University of Cambridge	99.2	99.1	98.8	99.1
3	University of Oxford	98.9	98.8	98.6	98.6
4	Harvard University	98.3	98.3	99.0	97.9
5	Stanford University	98.0	97.5	98.6	96.7

Introduction

Nowadays, higher education touches almost everyone's life. You might look at university rankings out of sheer curiosity—treating them as a global scoreboard for your dream school or alma mater. Alternatively, you might be a serious researcher or policymaker analyzing these metrics to uncover global educational trends and institutional strategies. Whichever group you belong to, you will find something intriguing in our visualization.

Our goals

Deconstruct the patterns of global excellence

We want to enable users to make their own conclusions and discover the patterns behind top-tier universities. By visualizing the data we have for over 200 institutions, we give the user the opportunity to interactively explore and make connections beyond a simple "overall score"—using their own logic to understand what truly makes a university great.

Visualize connections between ranking metrics

The key factors for a university's success are highly interconnected. We would like to visualize how different metrics (like Academic Reputation and Research Citations) correlate with one another, and how geographical regions exhibit distinct strengths across these indicators.

Create an engaging, editorial experience

Make the whole experience of exploring complex academic data engaging, interactive, and visually appealing. We aim to transform dense spreadsheets into an intuitive and immersive data storytelling dashboard.

What can you discover here?

Grasp the Big Picture: Looking for a high-level overview? Sit back and use our "Auto-Play" features to watch how academic dominance shifts across continents from 2023 to 2026. It's an elegant, effortless way to understand the geopolitical landscape of higher education.

Dive into Granular Details: Want to know exactly *why* a specific university ranks where it does? Dive into our interactive scatter plots and radar charts to conduct head-to-head benchmarking. You'll deconstruct the exact formula of indicators—from sustainability to research networks—that shape an institution's global standing.

The journey to the final result

Data Structuring & Base Visualization

Our initial step was making sense of four years of raw QS ranking data (2023–2026). Rather than just plotting numbers, we spent time meticulously cleaning and structuring the dataset so it could tell a cohesive global story. By the end of Milestone 2, we had successfully translated this data into the foundational skeletons of our core visual components: the Map, Scatter Plot, Radar Chart, and Ranking Table.

Meaningful Interactions & Cross-Component Discovery.

Building upon the static charts from M2, our primary goal was to transform isolated graphs into an interconnected exploration tool. We thought deeply about the user's path of discovery:

- **Contextual Filtering:** Clicking a country on the Map now seamlessly filters the dashboard to display only the ranked universities within that specific nation.
- **Fluid Exploration:** The Scatter Plot was upgraded to allow users to switch dynamically between different academic metrics, helping them visually uncover hidden correlations.
- **Deep Dives:** If a user spots an interesting outlier on the Scatter Plot, clicking that dot instantly scrolls them down to the exact row in the Ranking Table to investigate the raw data.
- **Flexible Benchmarking:** For the Radar Chart, we wanted to give users multiple ways to compare institutional DNA. They can either manually select universities from a dropdown menu or intuitively click the "Compare" buttons directly inside the Ranking Table to load them into the chart.

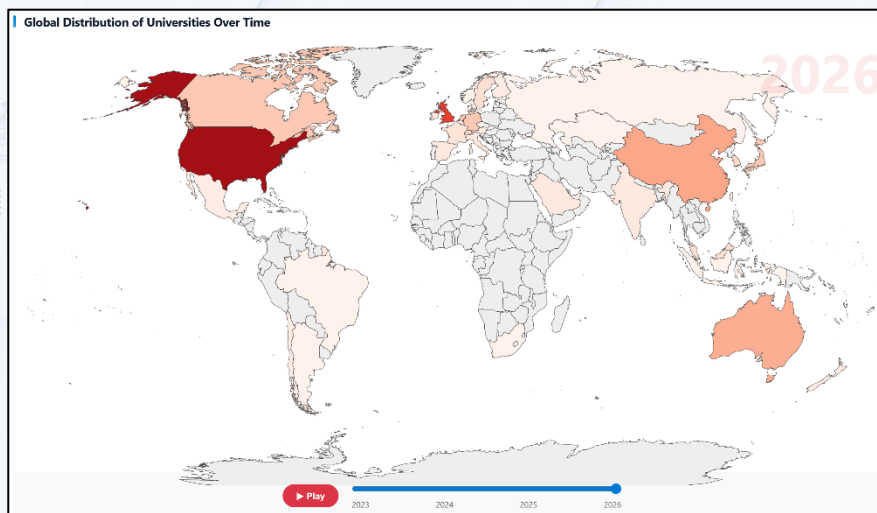
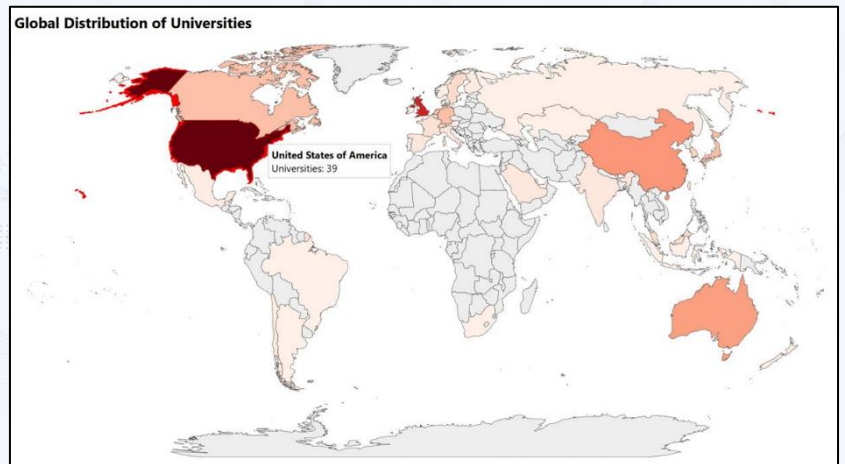
UI Design & Time-Based Storytelling.

In our final phase, we shifted our mindset from building a "functional dashboard" to crafting an "engaging editorial experience." We focused heavily on the overall UI design, organizing the layout to feel clean, modern, and accessible. More importantly, we introduced **animation over time**. Instead of forcing users to manually click through the years or metrics, we designed auto-playing timelines that visually animate the shifts in global rankings and metric distributions. We ensured that every transition was smooth, allowing the user's eye to naturally follow the data as it changes over time, turning raw statistics into a flowing narrative.

Evolution of Design

The global map

Initial Sketch: We envisioned a standard choropleth map. The color intensity would represent the number of top universities in each country. A simple timeline slider was planned to navigate through the years.



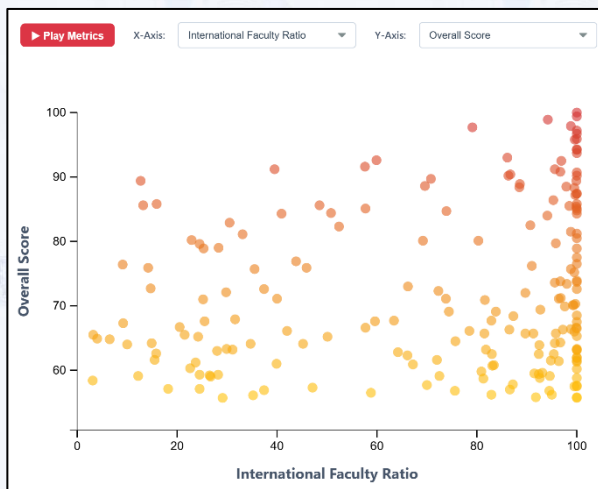
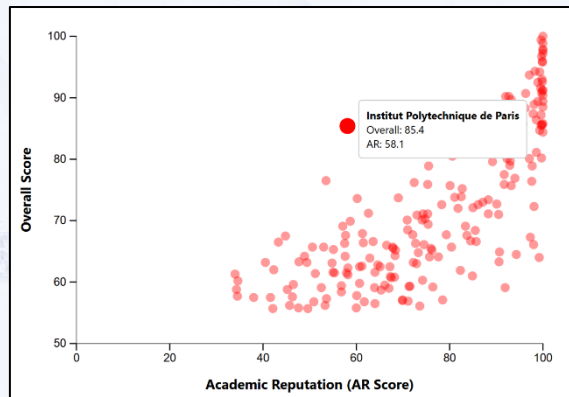
Evolution: The core concept remained, but the execution was significantly refined. Instead of a manual slider, we implemented an "Auto-Play" timeline, allowing users to watch the geographical shifts over four years dynamically. We also added detailed tooltips and ensured clicking a country filtered the entire dashboard.

United States of America - 2026 Total number (39)

Rank	University Name	Overall Score
1	Massachusetts Institute of Technology (MIT)	100.0
3	Stanford University	98.9
5	Harvard University	97.7
10	California Institute of Technology (Caltech)	94.3
13	University of Chicago	93.0
15	University of Pennsylvania	92.5
16	Cornell University	91.6
17	University of California, Berkeley (UCB)	91.2
21	Yale University	90.4
24	Johns Hopkins University	89.7
25	Princeton University	89.4
38	Columbia University	85.6

The Scatter Plot

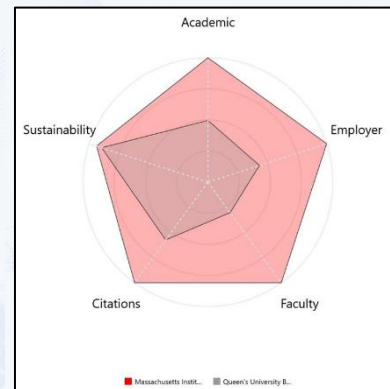
Initial Sketch: we designed a scatter plot, which is used to analyze relationships between ranking indicators and overall performance.



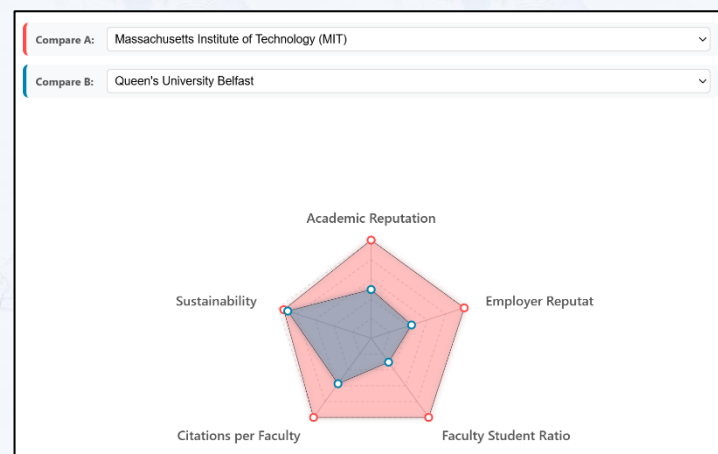
Evolution: We realized restricting the plot to just two metrics was limiting. We expanded this into an interactive tool where users can select any metric for the X and Y axes via dropdowns. The most significant addition was the "Play Metrics" feature, which loops through X-axis metrics automatically, revealing hidden correlations without requiring user input. We also refined the color gradient to map to the Y-axis score, shifting from yellow (lower) to deep red (higher).

The Radar Chart

Initial Sketch: We designed a radar chart to display the multi-dimensional profile of individual universities across core indicators.



Evolution: The design shifted to emphasize comparison. We integrated the radar chart closely with the ranking table. Users can use the dropdown menus to manually select universities or click "Red" or "Blue" buttons in the table to instantly load two universities into the radar chart.



The Ranking Table

Initial Sketch: The sketch showed a standard, paginated HTML table to display the raw ranking data, serving as a reference point.

Filter	Search Q							
Rank↑↓	Name	Regions	AR score↑↓	...				
1	MIT	USA	97	100				
2	...	USA	98	95				
3	...	UK	96	93				
4	...	UK	92	90				
5	...	China	90	91				

Evolution: The ranking table changed the most from our initial sketches. We realized a static, paginated table was too rigid, so we rebuilt it as a continuous, scrollable list. To enhance the visual quality of the dashboard, we replaced standard data snapping with smooth FLIP animations; rows now elegantly glide to their new ranks with a subtle red highlight to signify movement. We also added small "Compare" buttons inside the rows, creating a direct bridge between the raw data and the Radar Chart.

Detailed University Ranking Data

▶ Play

2026202520242023

Search university or country...

Show All Metrics ↑↓

RANK (#) ▲	UNIVERSITY NAME ↑↓	COUNTRY/REGION ↑↓	OVERALL SCORE ↑↓	COMPARE IN RADAR
1	Massachusetts Institute of Technology (MIT)	United States of America	100.0	RedBlue
2	Imperial College London	United Kingdom	99.4	RedBlue
3	Stanford University	United States of America	98.9	RedBlue
4	University of Oxford	United Kingdom	97.9	RedBlue
5	Harvard University	United States of America	97.7	RedBlue

Challenges

Challenge 1: Animating Table Ranks (The FLIP Animation)

When the user changes the year or sorts by metric, standard HTML tables simply snap to the new state. This caused users to lose track of whether a specific university moved up or down in the rankings.

Design Decision: Staggered FLIP + Highlights

We implemented a custom FLIP (First, Last, Invert, Play) animation for the table rows. When a row moves, it is assigned to a high z-index, a red highlight background, and a drop shadow. We applied a staggered delay ($i * 60\text{ms}$) to create a cascading "card shuffle" effect. Crucially, to save browser rendering resources, we restricted this complex animation to the Top 10 rows; the rest update instantly.

Challenge 2: Cross-Component Interruptions

We implemented auto-playback features for the map timeline, the ranking table, and the scatter plot. However, if a user clicked a dot on the scatter plot or on the global map to locate a university in the table, the ongoing animations would continue to trigger re-renders, causing the table to jump around and frustrate the user.

Design Decision: The "Interrupt & Sync" Architecture

We developed a global pause () API. Any cross-component interaction (like clicking a dot or clicking "Compare in radar") immediately fires a pause signal to all active timelines. Furthermore, we passed a skipAnimation flag to the table rendering engine during these cross-component clicks. This ensures the table instantly snaps to the correct data and flashes a yellow highlight on the targeted row, bypassing the 2.5-second FLIP animation for immediate feedback.

Peer Assessment & Work Breakdown

The development of the QS Explorer was a highly collaborative endeavor. While the core conceptualization, initial data cleaning, and cross-component integration logic were discussed and developed jointly as a team, individual members took ownership of specific visualization modules and design domains to ensure high-quality execution.

Shared Responsibilities: All team members participated in the initial brainstorming sessions, dataset selection, and the architectural planning of the dashboard. We jointly debugged the cross-filtering interactions (e.g., linking the map to the table) and collaboratively polished the final presentation.

Lei

- Global Map
- Ranking table
- Cross component interaction
- Screencast

Peiyu

- Scatter plot
- UI/UX Design
- Animation
- Cross component interaction

Yixiao

- Radar Chart
- UI/UX Design
- Cross component interaction
- Process Book

Resources

Our website: <https://com-480-data-visualization.github.io/CLZ/>

Our repository: <https://github.com/com-480-data-visualization/CLZ>

Our data: <https://www.topuniversities.com/>